Exercise 5
Frequency Response of Amplifiers

Winter Term 2025/26

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Exercise 5 – Bode plot

Each pole ω_p of the **transfer function** causes*

- a total phase shift of -90°
- a decay of the amplitude response of -20dB / decade
- a phase shift of -45° at $\omega = \omega_p$

Each zero ω_z of the **transfer function** causes*

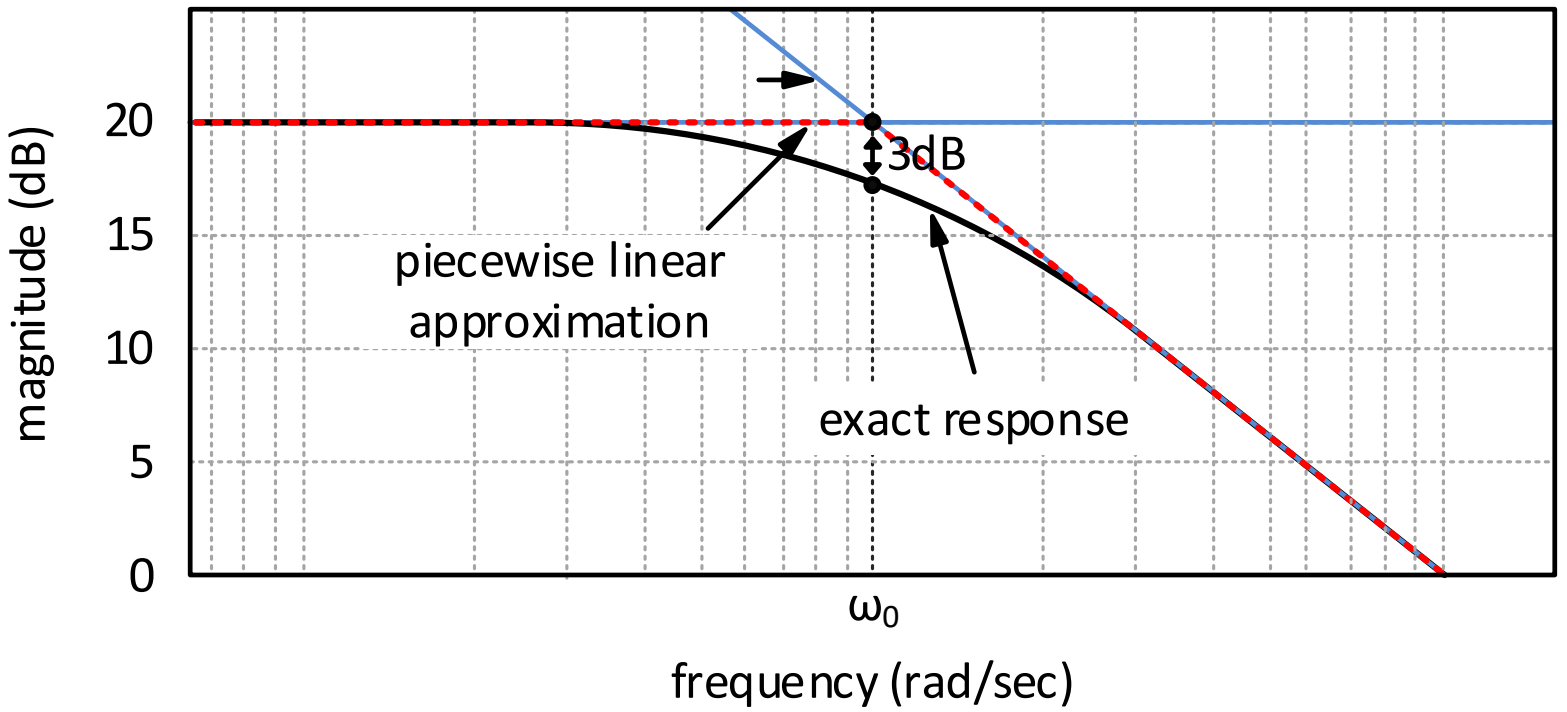
- a total phase shift of +90°
- an increase of the amplitude response of +20dB / decade
- a phase shift of +45° at $\omega = \omega_z$

The bode plot of a transfer functions with multiple poles/zeros results from the superposition of the bode plots of all single-pole and single-zero transfer functions.

*please note that this rules apply only for poles/zeros located in the negative half plane.

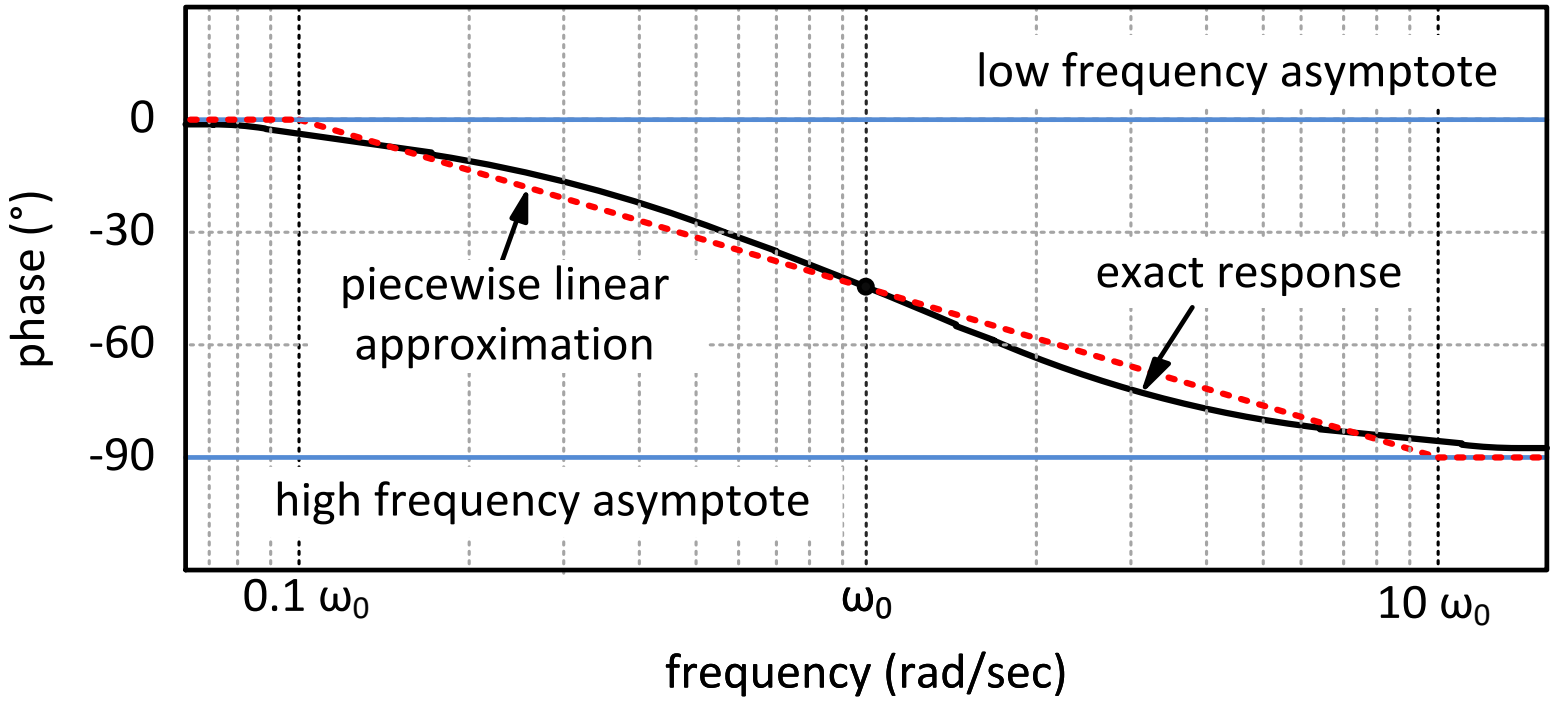
Exercise 5 – Bode plot

- **Piecewise linear approximation of magnitude response (pole at ω_0):**
  - Low frequency asymptote until ω_0
  - High frequency asymptote starts from ω_0



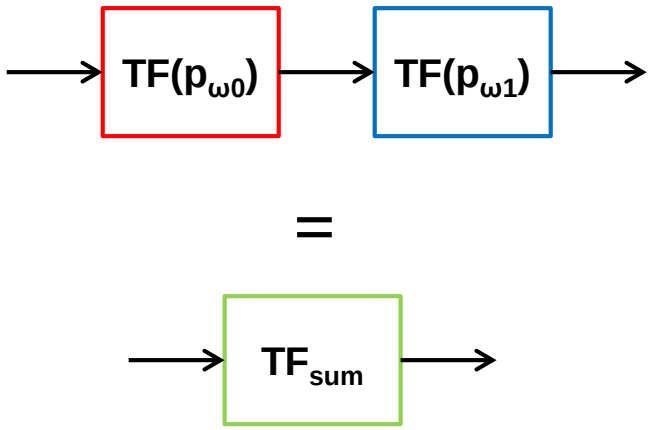
Exercise 5 – Bode plot

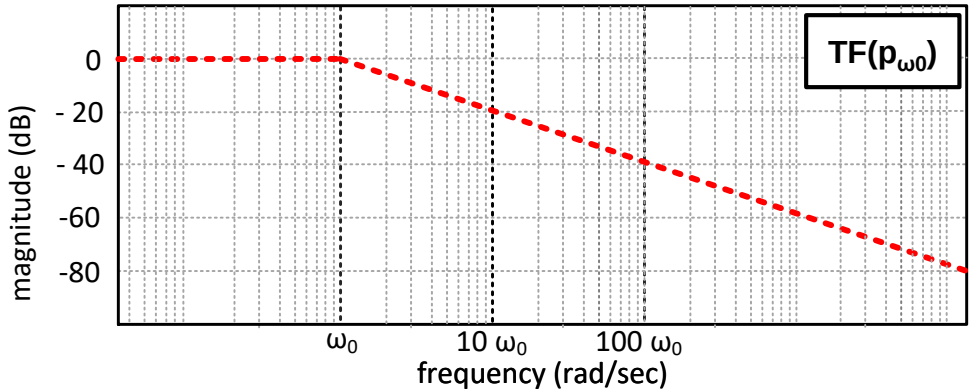
- **Piecewise linear approximation of phase response (pole at ω_0) :**
  - Low frequency asymptote until $\omega_0/10$
  - Phase decays linearly from $\omega_0/10$ to $10\omega_0$ (crossing the -45° point)
  - High frequency asymptote starts from $10\omega_0$

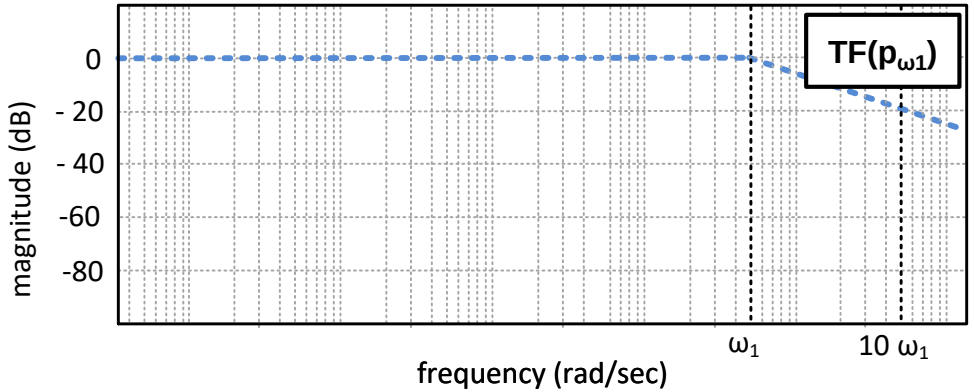


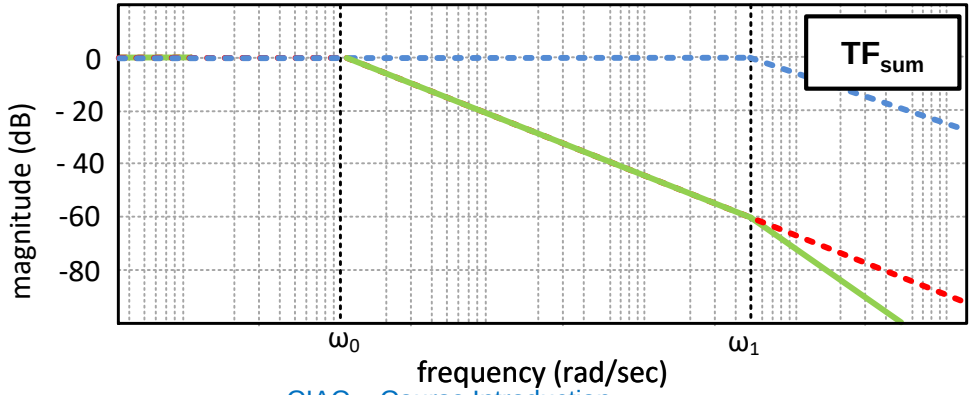
Exercise 5 – Bode plot

- System with
  2 real poles at ω_0 and ω_1
- Equivalent to the cascade of two single-pole systems
  → Magnitude responses add up



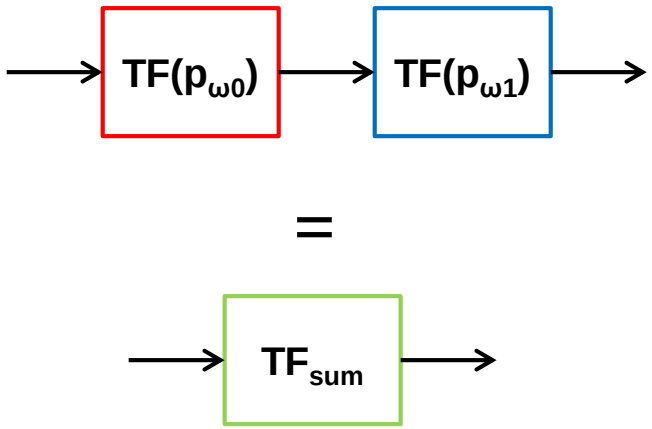


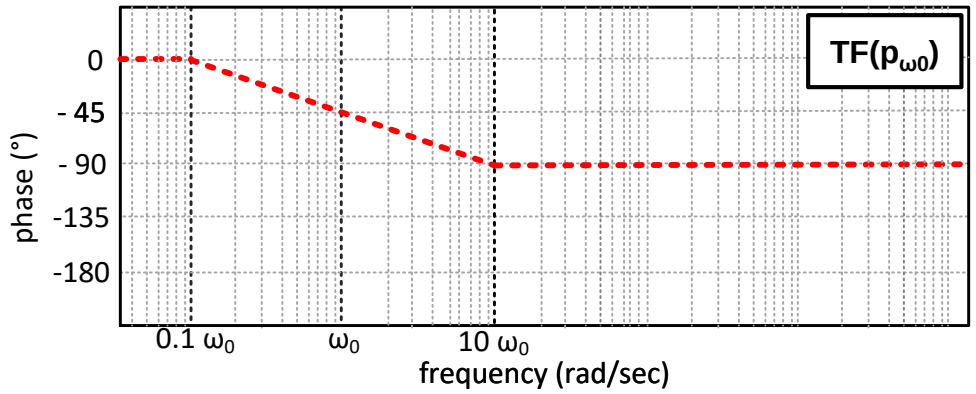


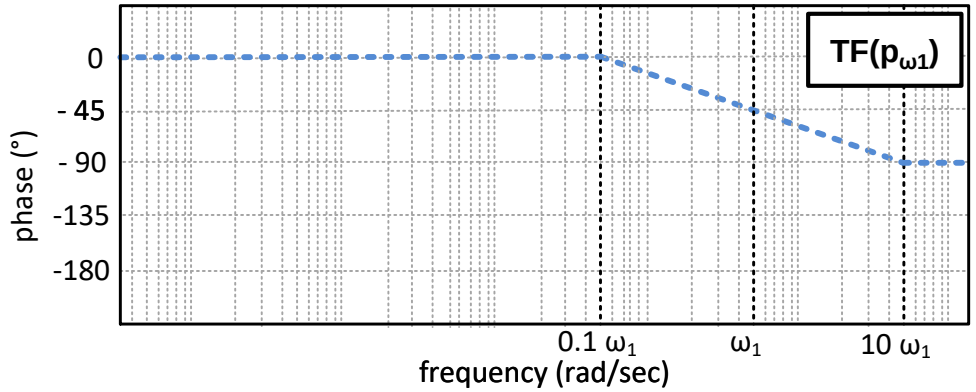


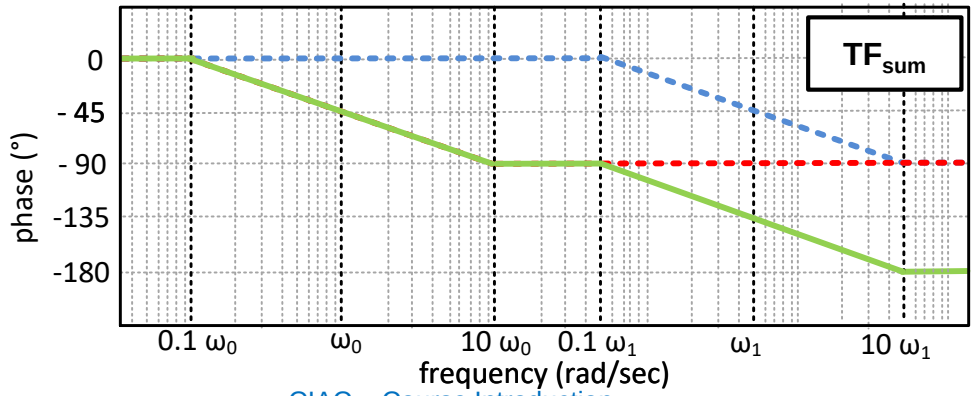
Exercise 5 – Bode plot

- System with
  2 real poles at ω_0 and ω_1
- Equivalent to the cascade of two single-pole systems
  → Phase responses add up



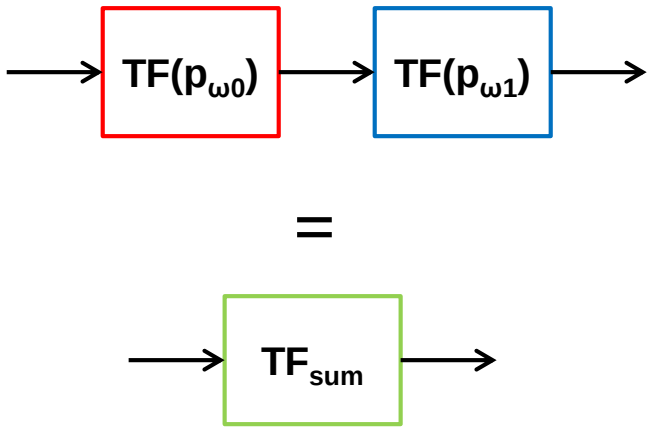


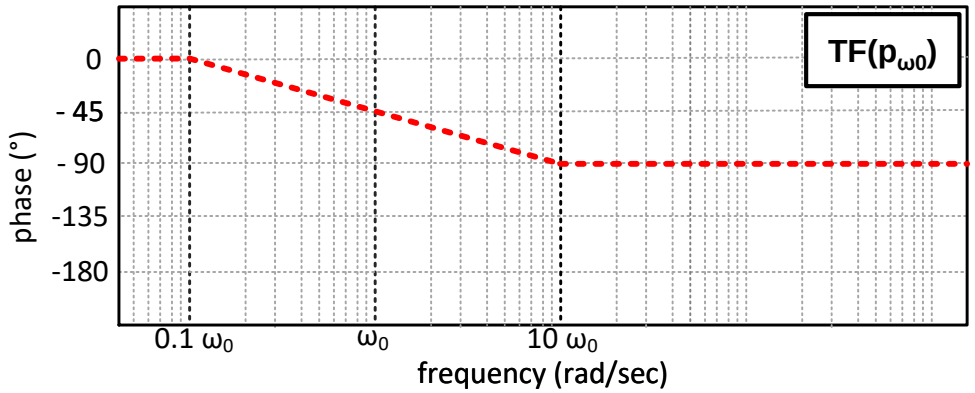


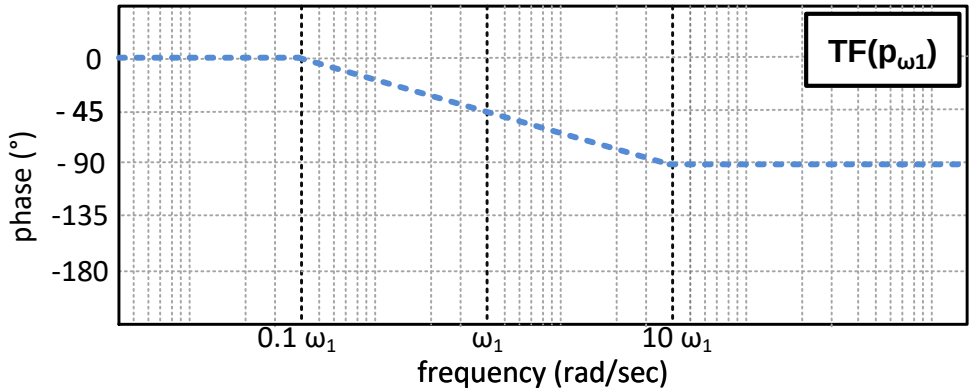


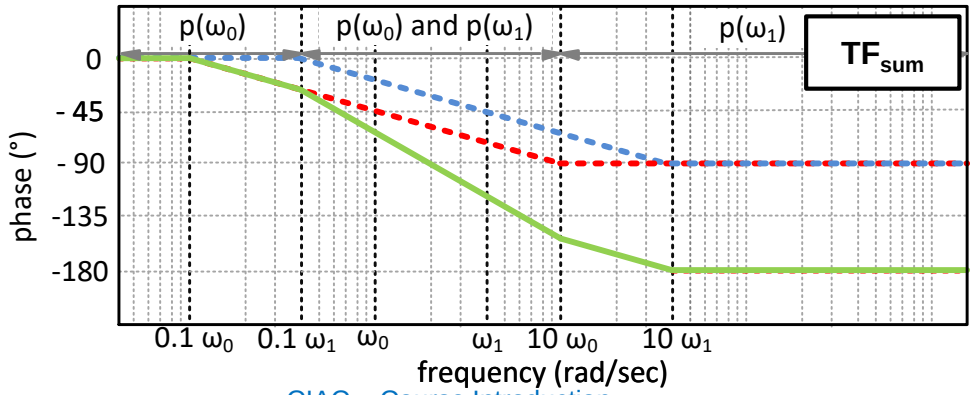
Exercise 5 – Bode plot

- System with
  2 real poles at ω_0 and ω_1
  (with overlapping responses)
- Equivalent to the cascade of two single-pole systems
  → Phase responses add up



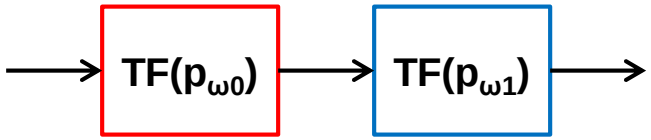






Exercise 5 – Bode plot

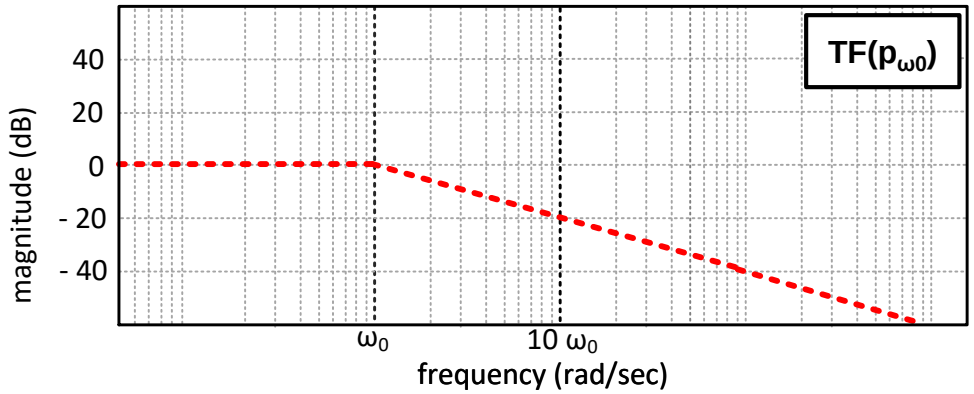
- System with
  1 real pole at ω_0 and
  1 real zero at ω_1
- Equivalent to the cascade of
  a single-pole and
  a „single-zero“ system*
  → Magnitude responses add up

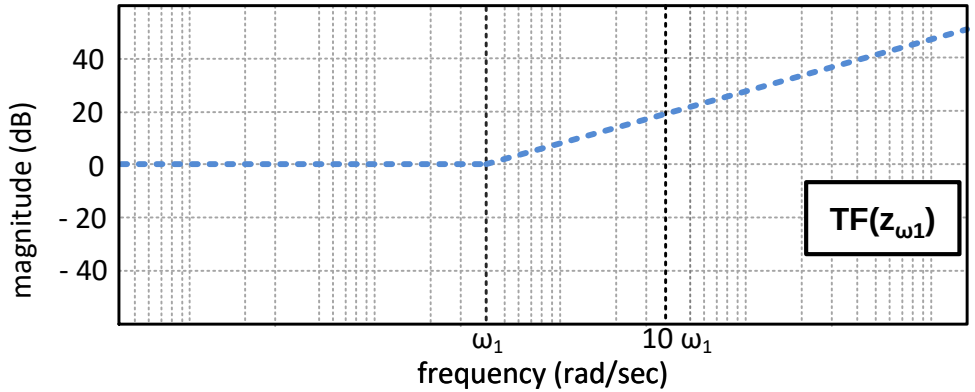


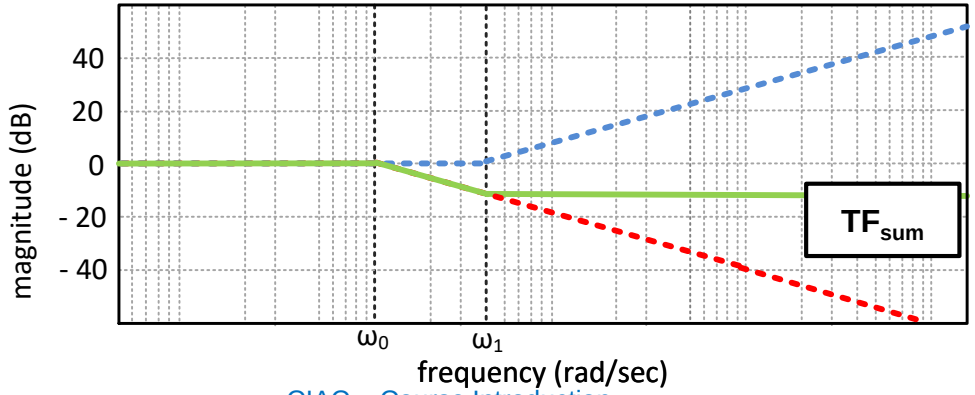
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*a „single-zero“ system is purely theoretical, no physical system can have more zeros than poles

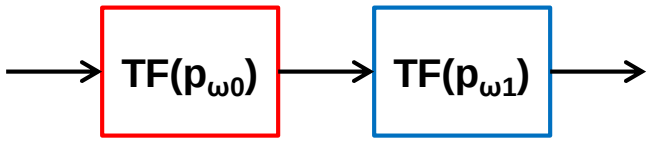






Exercise 5 – Bode plot

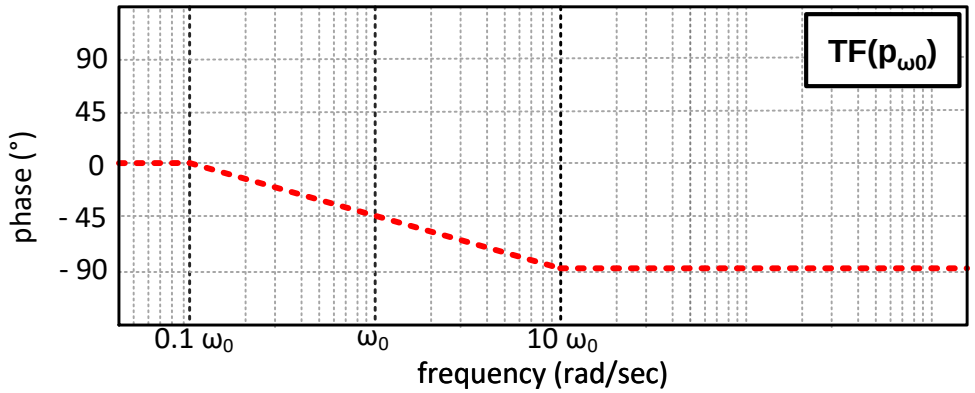
- System with
  1 real pole at ω_0 and
  1 real zero at ω_1
- Equivalent to the cascade of
  a single-pole and
  a „single-zero“ system*
  → Phase responses add up

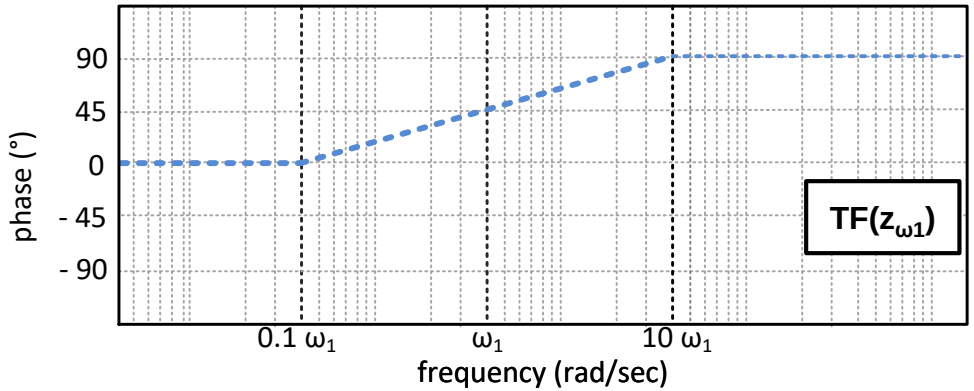


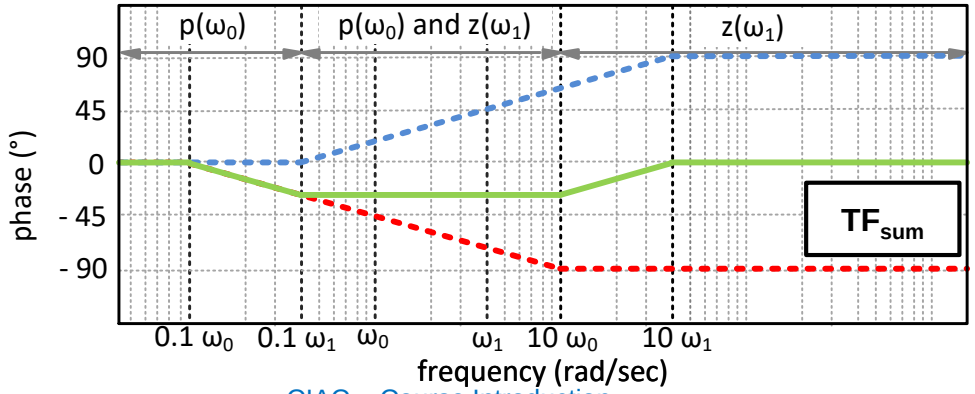
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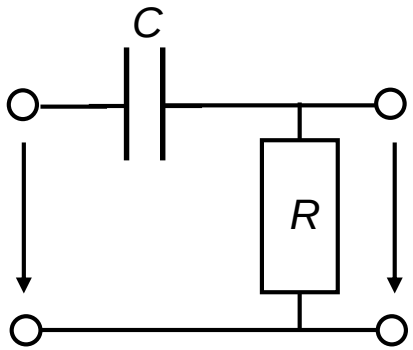






Exercise 5 – Task 1

1.1. Derive the transfer function $v_{\mathrm{OUT}}/v_{\mathrm{IN}}$ of the high-pass filter for R = 1MΩ and C = 1nF.



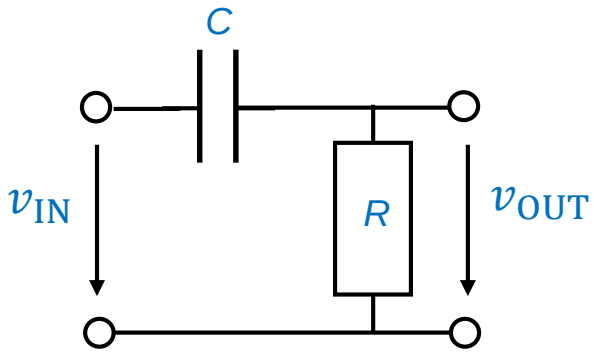
Exercise 5 – Task 1

1.1. Derive the transfer function $v_{\mathrm{OUT}}/v_{\mathrm{IN}}$ of the high-pass filter for R = 1MΩ and C = 1nF.

$$A(s) = \frac{v_{\mathrm{OUT}}(s)}{v_{\mathrm{IN}}(s)} = \frac{sRC}{1+sRC} \quad \xrightarrow{s = j\omega} \quad A(\omega) = \frac{v_{\mathrm{OUT}}(j\omega)}{v_{\mathrm{IN}}(j\omega)} = \frac{RCj\omega}{1+RCj\omega}$$

$$A(\omega) = \frac{j\omega RC}{1+j\omega RC} \cdot \frac{1-j\omega RC}{1-j\omega RC} = \frac{j\omega RC + C^2R^2\omega^2}{1+C^2R^2\omega^2}$$

$$A(\omega) = \frac{C^2R^2\omega^2}{1+C^2R^2\omega^2} + j\frac{\omega RC}{1+C^2R^2\omega^2} = (\mathrm{Re}) + j\,(\mathrm{Im})$$



$$|A(\omega)| = \left|\frac{v_{\mathrm{OUT}}(j\omega)}{v_{\mathrm{IN}}(j\omega)}\right| = \sqrt{(\mathrm{Re})^2 + (\mathrm{Im})^2} = \frac{RC\omega}{\sqrt{1+C^2R^2\omega^2}}$$

$$\phi(\omega) = \arctan\left(\frac{\mathrm{Im}}{\mathrm{Re}}\right) \quad \Rightarrow \quad \phi(\omega) = \arctan\left(\frac{1}{\omega RC}\right)$$

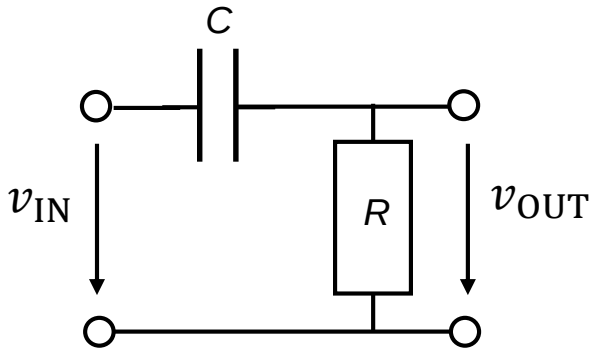
Exercise 5 – Task 1

1.2. Calculate the -3dB cut-off frequency f_c, i.e., the frequency for which the magnitude of the transfer function $v_{\mathrm{OUT}}/v_{\mathrm{IN}}$ results in 1/√2.

$$|A(\omega_c)| = \frac{RC\omega_c}{\sqrt{1+C^2R^2{\omega_c}^2}} = \frac{1}{\sqrt{2}} \rightarrow \quad \omega_c = \frac{1}{RC} \quad f_c = \frac{1}{2\pi RC}$$

For R= 1 MΩ, C= 1 nF :

$$\omega_c = 1\ \mathrm{k}\frac{\mathrm{rad}}{\mathrm{s}}$$
$$f_c = 159.155\ \mathrm{Hz}$$



$$\phi(\omega_c) = \arctan\left(\frac{1}{\omega_c RC}\right) = \arctan(1) = 45^\circ$$

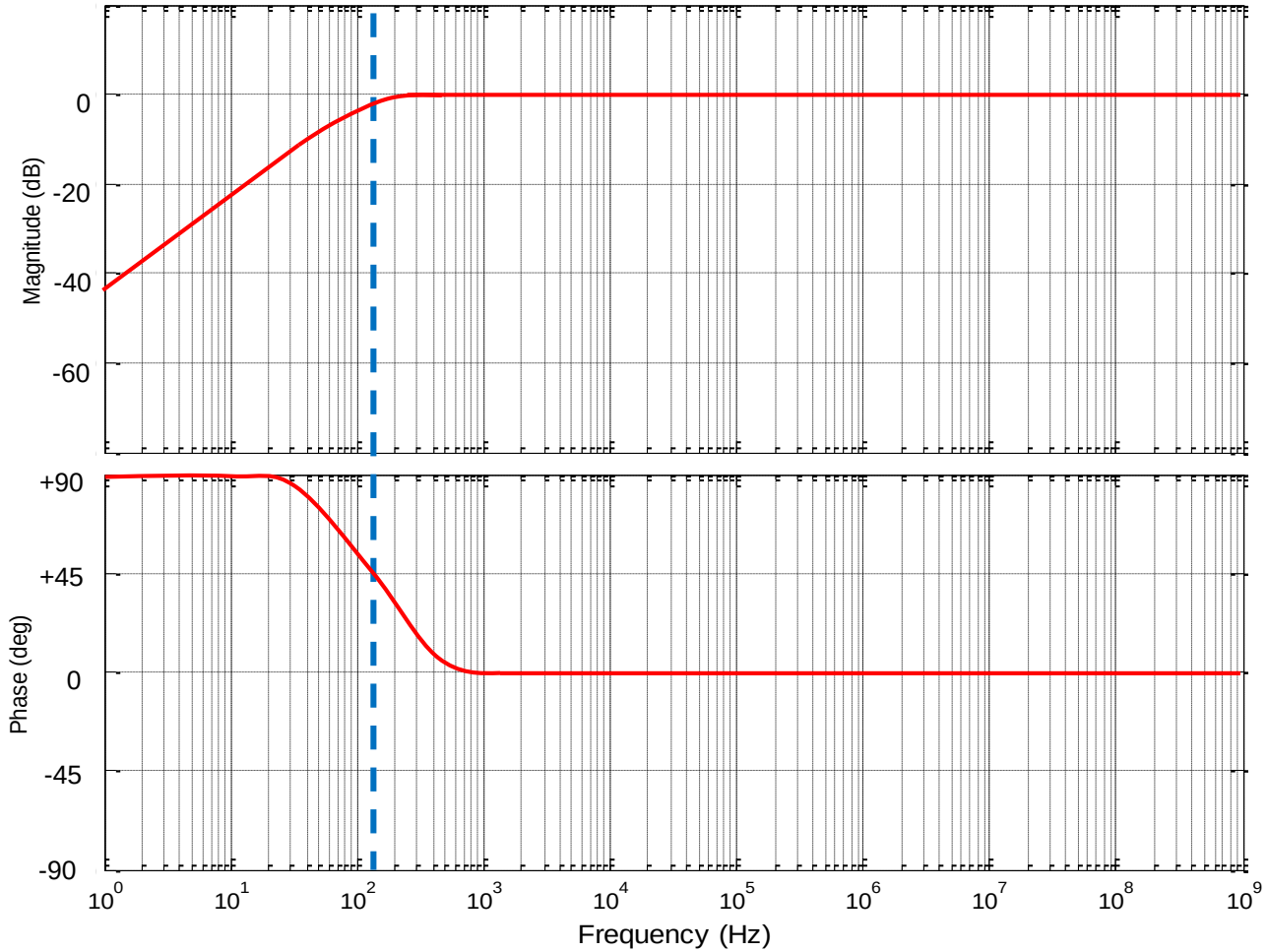
Exercise 5 – Task 1

1.3. Calculate the magnitude (in dB) and the phase (in degrees) of the high-pass filter for the following frequencies: $f = 0.01f_c$, $f = 0.1f_c$, $f = 0.5f_c$, $f = f_c$, $f = 2f_c$, $f = 10f_c$, and $f = 100f_c$!

f / f_c	phase (°)	magnitude (dB)
0.01	89.43	-40.00
0.10	84.29	-20.04
0.50	63.43	-6.99
1.00	45	-3.01
2.00	26.57	-0.97
10.00	5.71	-0.04
100.00	0.57	0

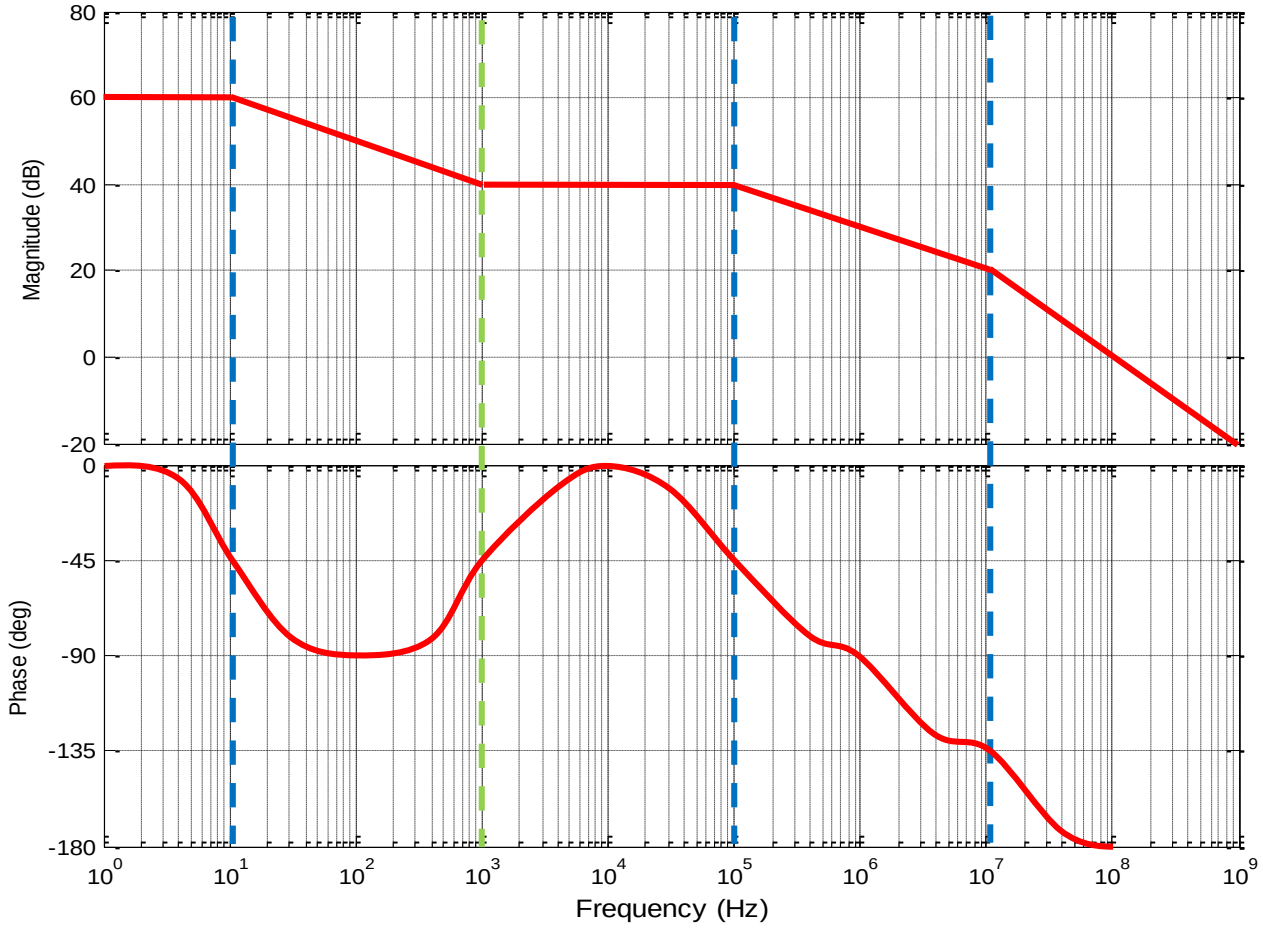
Exercise 5 – Task 1

1.4. Draw the Bode plot for R = 1MΩ and C = 1nF into the draft of an empty Bode plot on page 4.



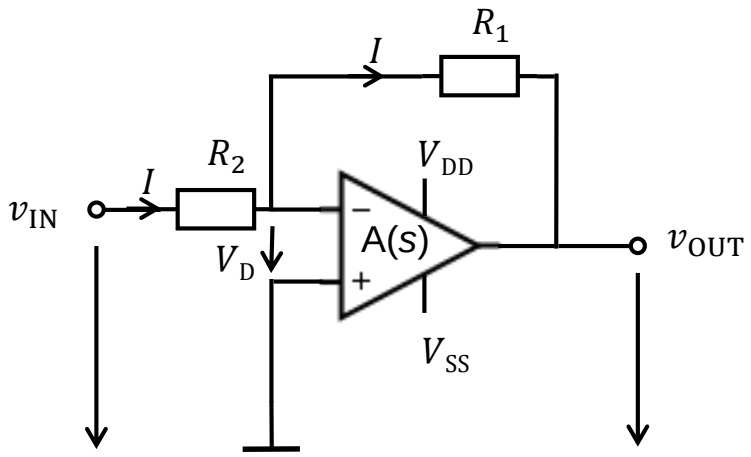
Exercise 5 – Task 1

1.5 Draw the Bode plot of an amplifier with the following parameters into the draft of an empty Bode plot on page 5: DC-gain: A_{DC} = 80 dB, poles: p_1 = 10 Hz, p_2 = 100 kHz, p_3 = 10 MHz, zeros: z_1 = 1 kHz



Task II

Stability of closed-loop systems



Exercise 5 – Task 2

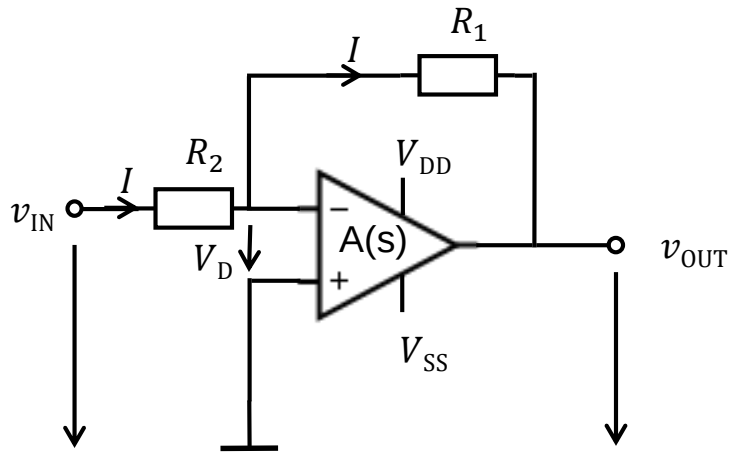
2.1. Derive the closed-loop transfer function $v_{\mathrm{OUT}}/v_{\mathrm{IN}}$ of the inverting amplifier as a function of A(s)!

$$v_{\mathrm{OUT}} = -A(s)V_{\mathrm{D}}$$

$$I = \frac{v_{\mathrm{IN}} - V_{\mathrm{D}}}{R_2} = \frac{V_D - v_{\mathrm{OUT}}}{R_1}$$

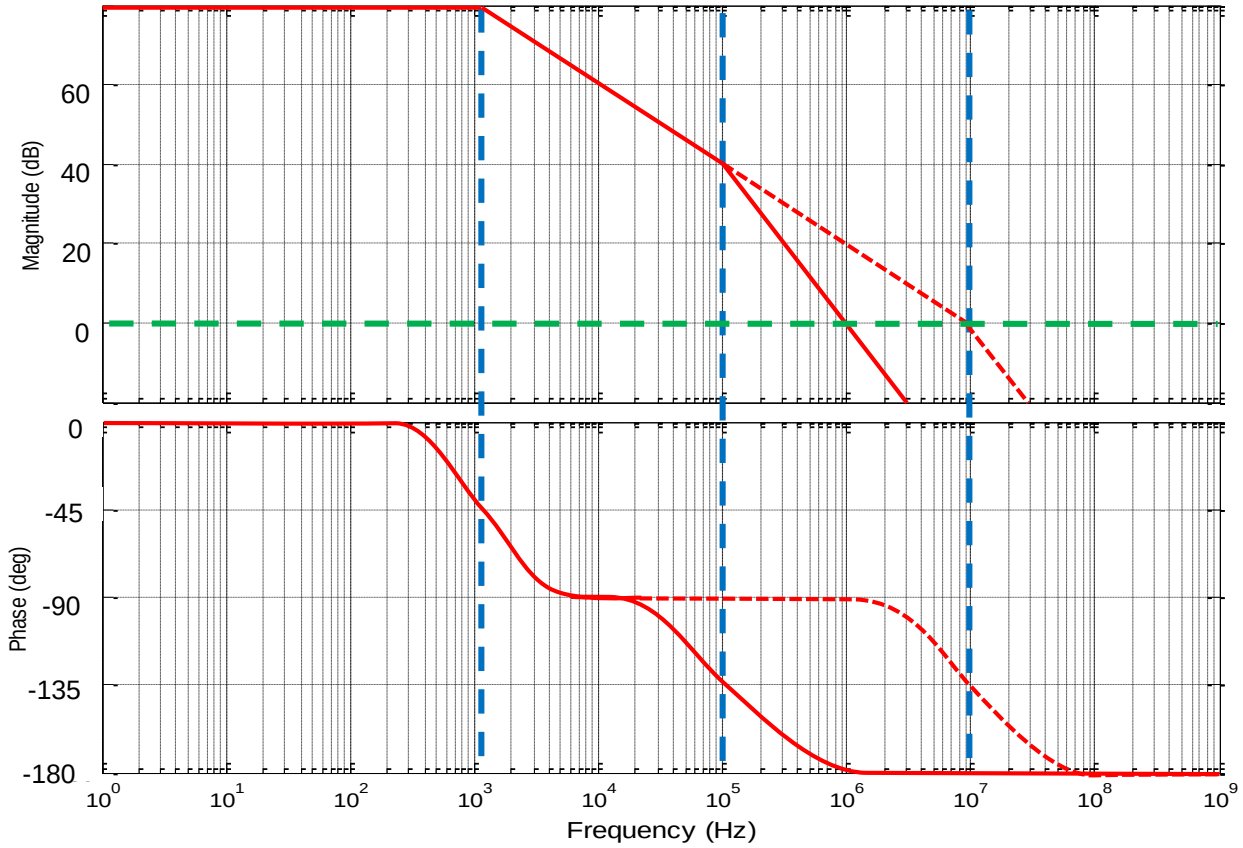
$$v_{\mathrm{OUT}} = -A(s)\frac{v_{\mathrm{OUT}} + v_{\mathrm{IN}}\frac{R_1}{R_2}}{1+\frac{R_1}{R_2}}$$

$$\frac{v_{\mathrm{OUT}}}{v_{\mathrm{IN}}} = -\frac{\frac{R_1}{R_2}}{1+\frac{1}{A(s)}\left(1+\frac{R_1}{R_2}\right)}$$



Exercise 5 – Task 2

2.2. The transfer function $A(s)$ of the amplifier can be modelled as a single-pole system with A_{DC} = 80dB and p_1 = 1 kHz. Sketch the Bode plot of $A(s)$ assuming a second pole p_2 for the following cases: p_2 = 100 kHz and p_2 = 10MHz!

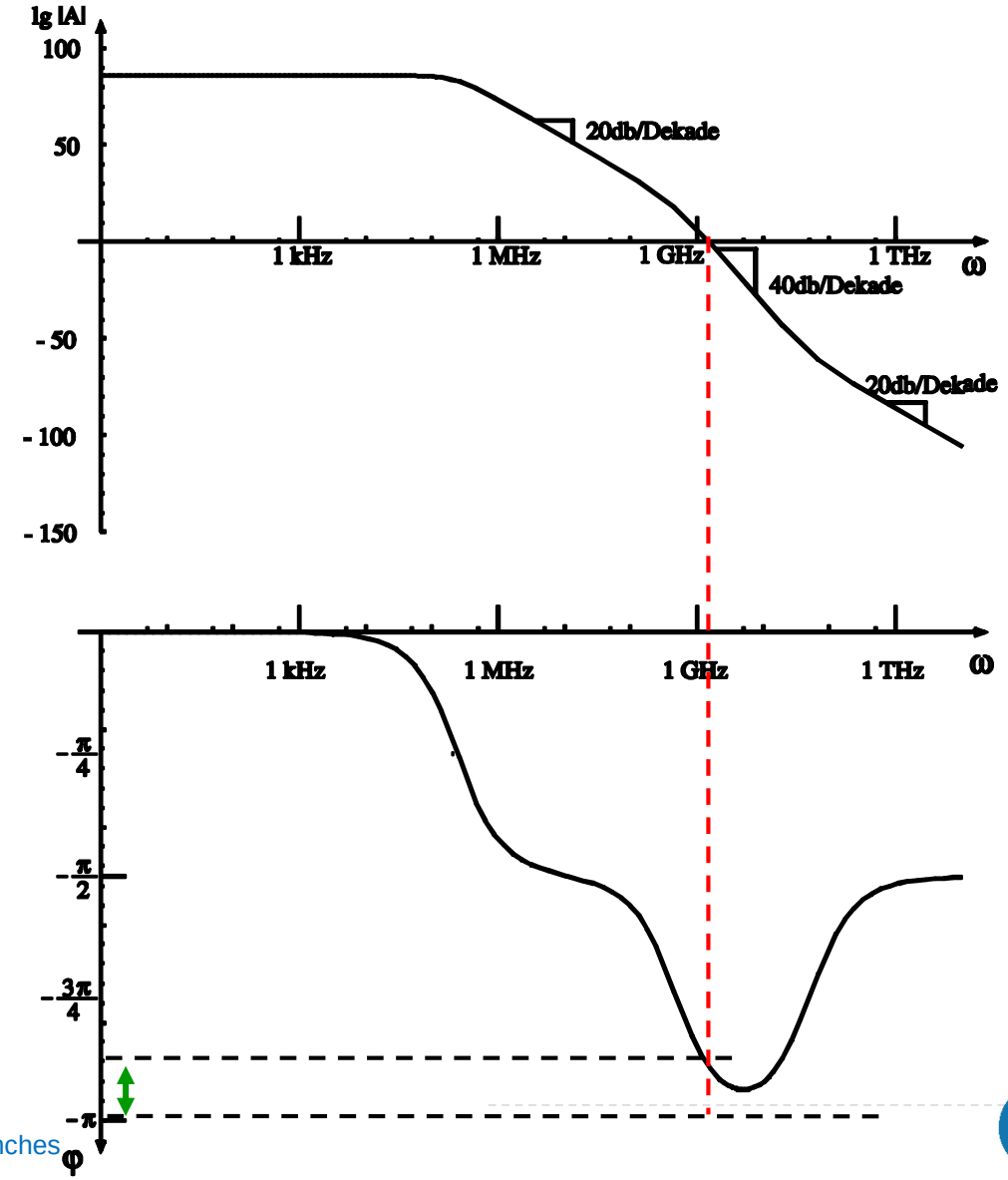


Exercise 5 – Task 2

Phase margin:

If $|H_0(j\omega)| = 1$, i.e., 0dB

→ Which phase difference do we have to -180°?



Exercise 5 – Task 2

2.3. Determine the gain and phase margin for the cases considered in task 2.2 and sketch the corresponding step responses in the time domain!

$A_0 = 80\text{dB} = 10000, \boldsymbol{\omega_{p1} = 2\pi\ 1\text{kHz}, \omega_{p2} = 2\pi\ 100\text{kHz}}$

$$A(s) \approx A_0 \frac{1}{\frac{s}{\omega_{p1}} + 1} \frac{1}{\frac{s}{\omega_{p2}} + 1}$$

$$\phi(\omega) = \arctan\left(\frac{Im}{Re}\right) = -\arctan\left(\frac{\omega}{\omega_{p1}}\right) - \arctan\left(\frac{\omega}{\omega_{p2}}\right)$$

$\omega_u = 2\pi \cdot 1\text{MHz}$ (from Bode plot)

$$\phi(\omega_u) = -\arctan\left(\frac{1\ \text{MHz}}{1\text{kHz}}\right) - \arctan\left(\frac{1\text{MHz}}{100\text{kHz}}\right)$$

phase margin $\approx 180° - \phi(\omega_u) \approx 5.7°$

Gain margin is infinite.
The phase shift of a two-pole system reaches -180° for $\omega \rightarrow \infty$, where the gain approaches - ∞.

Exercise 5 – Task 2

2.3. Determine the gain and phase margin for the cases considered in task 2.2 and sketch the corresponding step responses in the time domain!

$$A_0 = 80\text{dB} = 10000, \omega_{p1} = 2\pi\ 1\text{kHz}, \boldsymbol{\omega_{p2} = 2\pi\ 10\text{MHz}}$$

$$A(s) \approx A_0 \frac{1}{\frac{s}{\omega_{p1}} + 1} \frac{1}{\frac{s}{\omega_{p2}} + 1}$$

$$\phi(\omega) = \arctan\left(\frac{Im}{Re}\right) = -\arctan\left(\frac{\omega}{\omega_{p1}}\right) - \arctan\left(\frac{\omega}{\omega_{p2}}\right)$$

$$\omega_u = 2\pi \cdot 10\text{MHz} \quad \text{(from Bode plot)}$$

$$\phi(\omega_u) = -\arctan\left(\frac{10\ \text{MHz}}{1\text{kHz}}\right) - \arctan\left(\frac{10\text{MHz}}{10\text{MHz}}\right)$$

phase at ω_{p2}

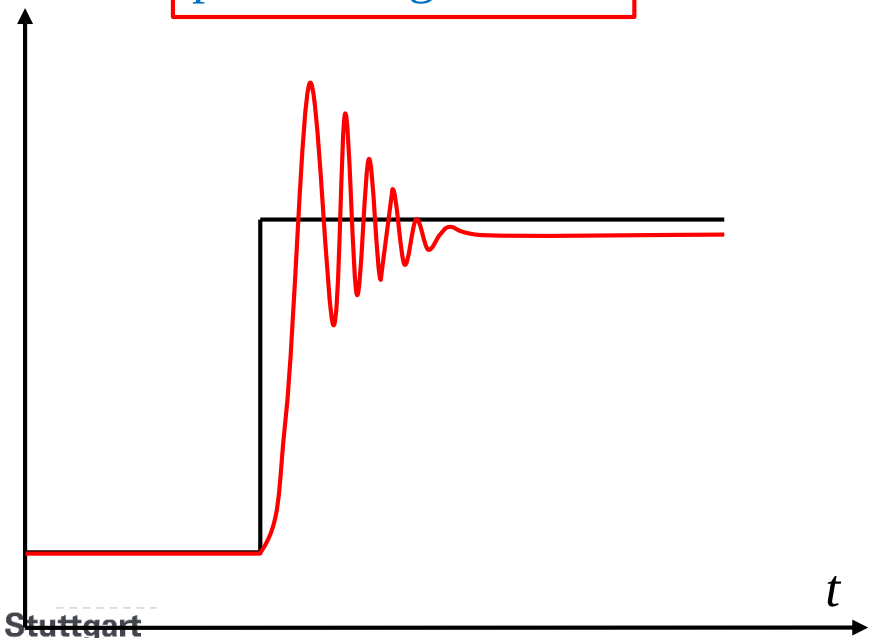
$$\text{phase margin} \approx 180° - \phi(\omega_u) \approx 44.9°$$

Gain margin is infinite.
The phase shift of a two-pole system reaches -180° for $\omega \rightarrow \infty$, where the gain approaches - ∞.

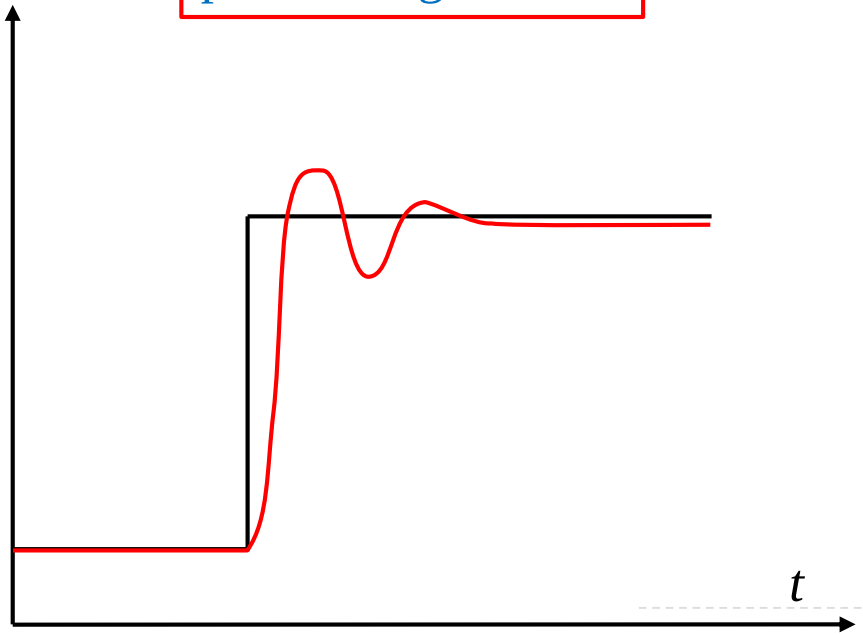
Exercise 5 – Task 2

3.3. Determine the gain and phase margin for the cases considered in task 2.2 and sketch the corresponding step responses in the time domain!

A_0 = 80 dB =10000
f_{p1} = 1 kHz
f_{p2} = 100 kHz
phase margin: 5.7°

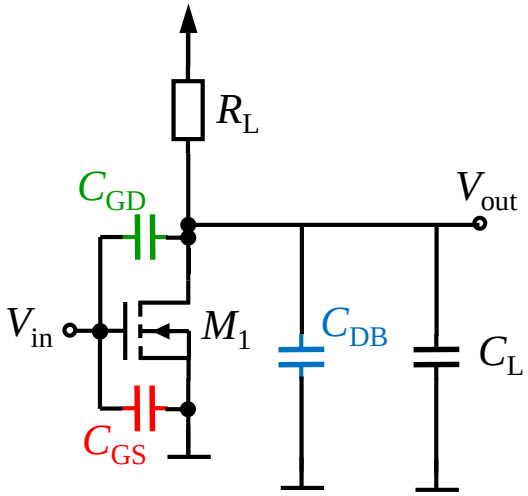


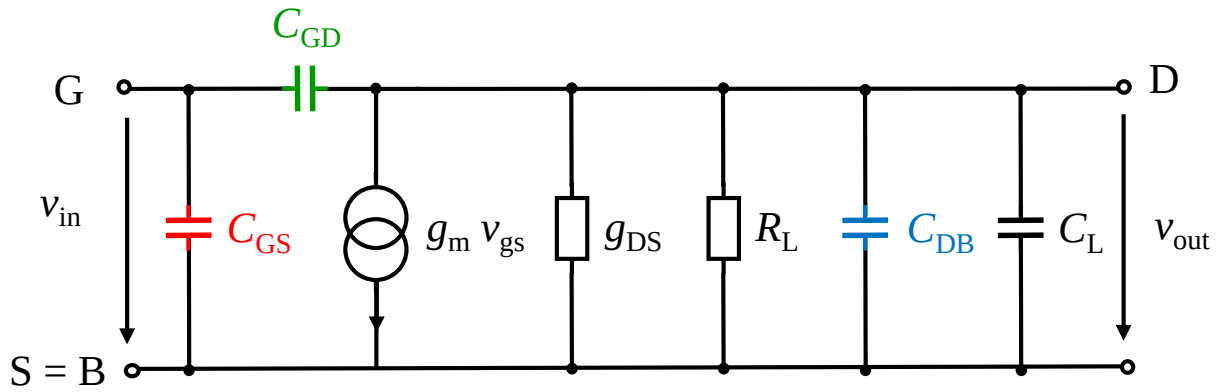
A_0 = 80 dB =10000
f_{p1} = 1 kHz
f_{p2} = 10 MHz
phase margin: 45°



Exercise 5 – Task 3

3.1. Draw the small-signal equivalent circuit! Based on the approach to the determination of the Y-parameters, derive an equation for the small-signal voltage gain. The load capacitor and the parasitic capacitors are to be considered!

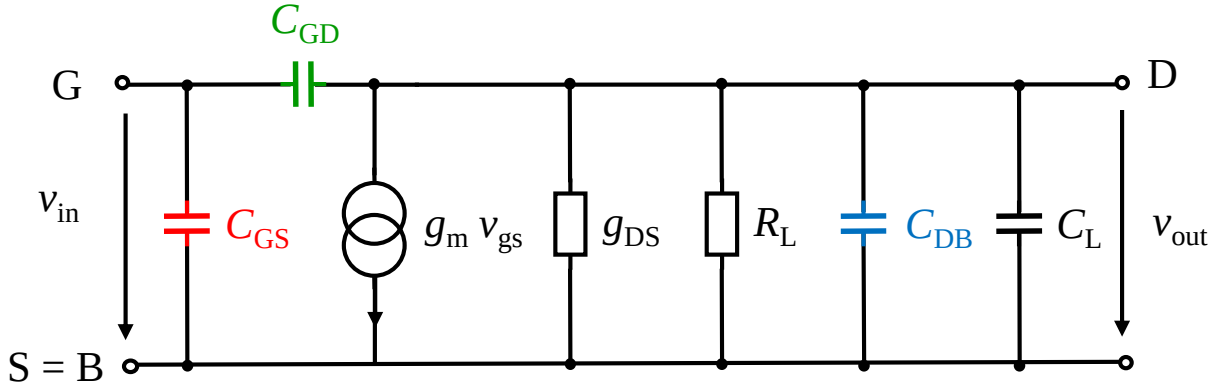


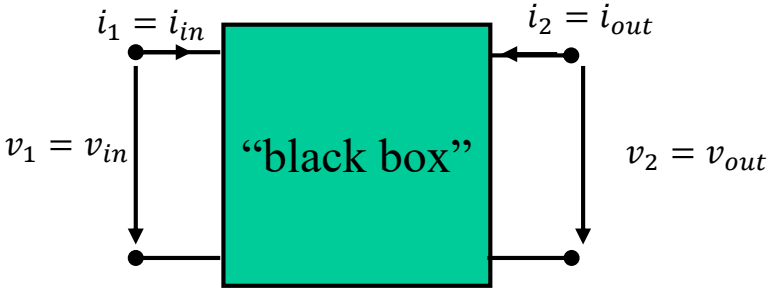


Transfer function: $A_v(s) = - G_m z_{out}$

Exercise 5 – Task 3

3.1. Draw the small-signal equivalent circuit! Based on the approach to the determination of the Y-parameters, derive an equation for the small-signal voltage gain. The load capacitor and the parasitic capacitors are to be considered!





$$A_v(s) = - G_m z_{out}$$

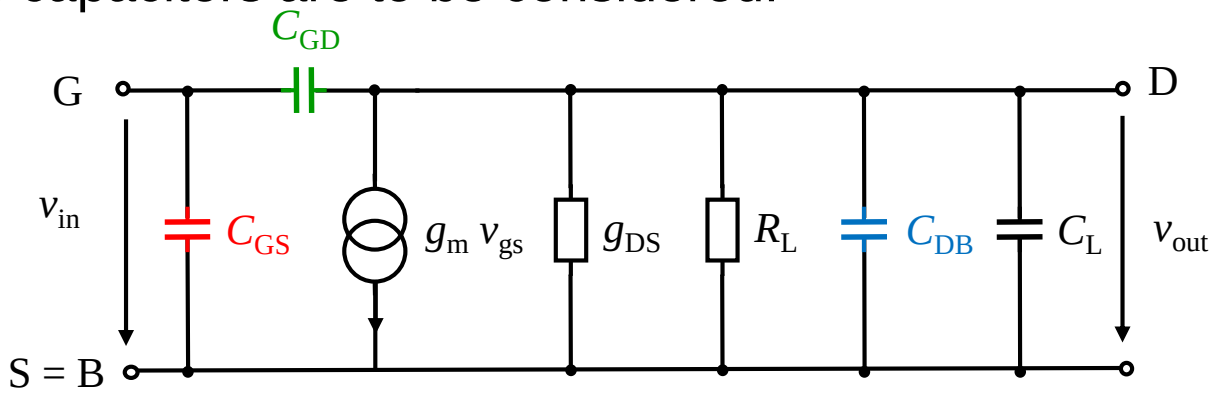
$$i_{out} = Y_{21} v_{in} + Y_{22} v_{out}$$

$$G_m = Y_{21} = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0} \qquad Y_{out} = Y_{22} = \left.\frac{i_{out}}{v_{out}}\right|_{v_{in}=0}$$

$$A_v(s) = -G_m z_{out} = - \frac{Y_{21}}{Y_{22}}$$

Exercise 5 – Task 3

4.1. Draw the small-signal equivalent circuit! Based on the approach to the determination of the Y-parameters, derive an equation for the small-signal voltage gain. The load capacitor and the parasitic capacitors are to be considered!



$$G_m = Y_{21} = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0} = g_m - sC_{GD}$$

$$Y_{out} = Y_{22} = \left.\frac{i_{out}}{v_{out}}\right|_{v_{in}=0}$$

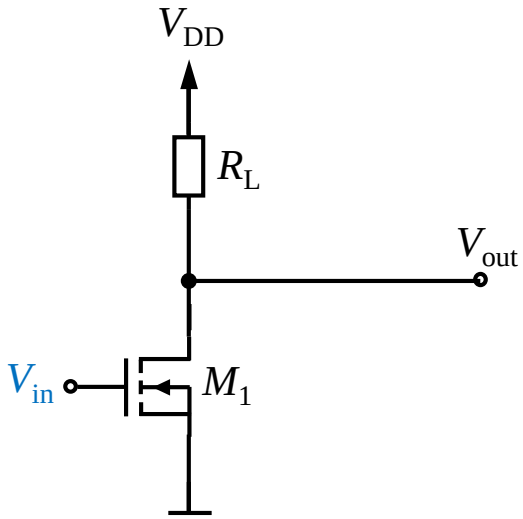
$$z_{out} = \frac{1}{Y_{22}} = \frac{r_{DS} \| R_L}{1 + s(C_{GD} + C_{DB})(r_{DS} \| R_L)}$$

$$A_v(s) = -G_m z_{out} = -(g_m - sC_{GD}) \frac{r_{DS} \| R_L}{1 + s(C_{GD} + C_{DB})(r_{DS} \| R_L)}$$

Exercise 5 – Task 3

3.2. Calculate the small signal DC-gain, i.e., ω = 0, as well as all poles and zeros using the following parameters:

V_{DD} = 1.2 V	V_{in} = 0.35 V	W = 20 µm	L= 10 µm
β_o = 400 µA/V^2	V_T = 200 mV	C_{GDov0} = 400 pF/m	C_{DB} can be neglected
R_L = 50 kΩ	C_L=1 pF	λ = 0.02 V^{-1}	



large-signal behavior (M_1 in saturation)

$$I_D = \frac{\beta}{2}(V_{GS} - V_T)^2$$

$$V_{out} = V_{DS} = V_{DD} - I_D R$$

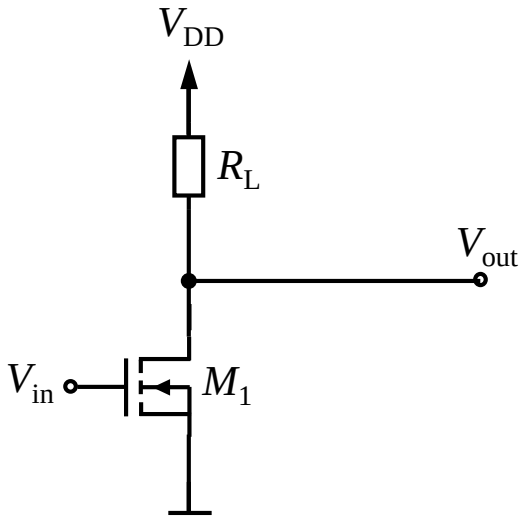
I = 9 µA
V_{out} = 0.75 V

$V_{DS} > (V_{GS} - V_T)$ → assumption “saturation” correct

Exercise 5 – Task 3

3.2. Calculate the small signal DC-gain, i.e., ω = 0, as well as all poles and zeros using the following parameters:

V_{DD} = 1.2 V	V_{in} = 0.35 V	W = 20 µm	L= 10 µm
β_o = 400 µA/V^2	V_T = 200 mV	C_{GDov0} = 400 pF/m	C_{DB} can be neglected
R_L = 50 kΩ	C_L=1 pF	λ = 0.02 V^{-1}	



small-signal voltage gain

$$A_v = -g_m r_{out}$$

$$g_m \approx \beta\,(U_{GS} - U_T) \approx 120\ \mu S$$

$$r_{DS} = \frac{1}{\lambda\, I_D} = 5.56\ M\Omega$$

$$r_{out} = r_{DS} || R_L = 49\ k\Omega$$

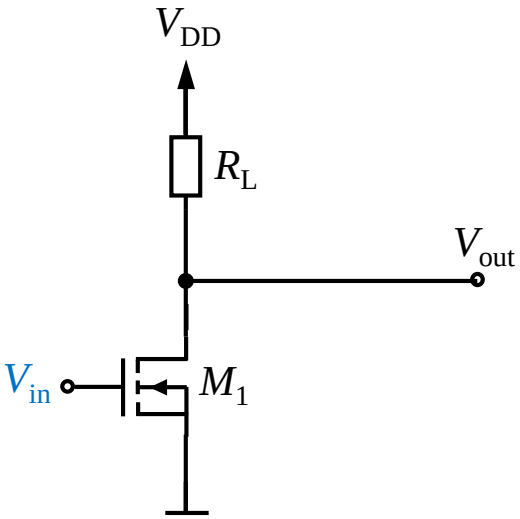
A_V = - 5.88

→ 15.39 dB

Exercise 5 – Task 3

3.2. Calculate the small signal DC-gain, i.e., ω = 0, as well as all poles and zeros using the following parameters:

V_{DD} = 1.2 V	V_{in} = 0.35 V	W = 20 µm	L= 10 µm
β_o = 400 µA/V^2	V_T = 200 mV	C_{GDov0} = 400 pF/m	C_{DB} can be neglected
R_L = 50 kΩ	C_L=1 pF	λ = 0.02 V^{-1}	



Zeros:

$$A_v(s) = -(g_m - sC_{GD})\frac{r_{DS}\|R_L}{1 + s(C_{GD} + C_{DB})(r_{DS}\|R_L)}$$

$$s_z = +\frac{g_m}{C_{GD}}$$

$C_{GD} = C_{GDov}$ (saturation, see chapter 1)

$= C_{GDov0} \cdot W$

$= 8$ fF

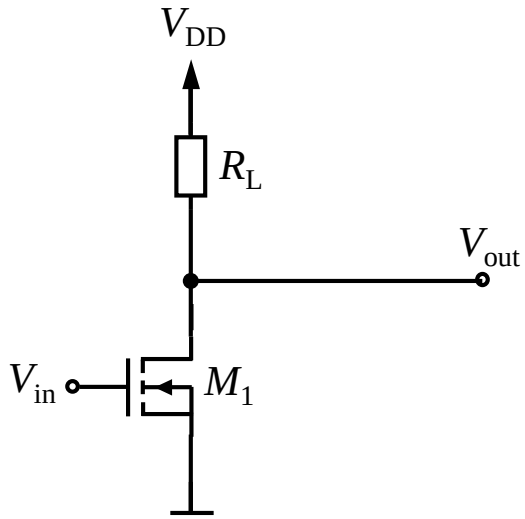
$s_z = 15$ Grad/s

$f_z = 2.39$ GHz

Exercise 5 – Task 3

3.2. Calculate the small signal DC-gain, i.e., ω = 0, as well as all poles and zeros using the following parameters:

V_{DD} = 1.2 V	V_{in} = 0.35 V	W = 20 µm	L= 10 µm
β_o = 400 µA/V^2	V_T = 200 mV	C_{GDov0} = 400 pF/m	C_{DB} can be neglected
R_L = 50 kΩ	C_L=1 pF	λ = 0.02 V^{-1}	



Poles:

$$A_v(s) = -(g_m - sC_{GD})\frac{r_{DS}\|R_L}{1 + s(C_{GD} + C_{DB})(r_{DS}\|R_L)}$$

$$s_p = -\frac{1}{(r_{DS}\|R_L)(C_{GD} + C_{DB} + C_L)}$$

s_p = 20 Grad/s

f_p = 3.22 GHz

Exercise 5 – Task 3

3.4. From the simulation results, determine the DC-gain, the poles, and the zero and compare them with the results of your calculation!

	small signal gain $\|A_V\|$		zero s_z		pole s_p	
	V/V	dB	Grad/s	GHz	Mrad/s	MHz
calculated	5.88	15.39	15.00	2.39	20.00	3.22
simulated	5.79	15.25	13.3	2.12	19.96	3.12